Evidence-Gated Crop-Loss Claim Verification and Profile-Aware Advisory Chat for Smallholder Farmers: Design, Conformance Evaluation and Live Feasibility Study

Author names withheld in this draft

Abstract*—Smallholder crop-loss claims are hard to verify: photographs, a land document, a polygon drawn on a map and a satellite vegetation index each say something different, and none alone establishes who may claim or whether a loss occurred. We describe Vivasayi AI, a farmer-facing system for Tamil Nadu, India, that pairs a profile-aware bilingual advisory chat with a claim-verification pipeline in which every evidence source is reduced to an explicit per-component outcome (verified, failed, unverified, unavailable, inconclusive, pending, requires review) and a deterministic, versioned rule engine, never a language model, decides. A required-component gate makes approval depend on claimant-rights evidence, holds a claim instead of deciding when evidence is pending, and never reads a provider failure as a negative finding. On deterministic synthetic benchmarks with analytic or specification-based ground truth, the legacy photograph-only policy approved 10 of 190 scenarios that the design rule forbids, and the gate approved none. In a live feasibility study, the production Sentinel-2 step ran end to end on 12 parcel-sized areas: near three documented 2023–2024 flood and cyclone events it reported no supporting loss in any of 9 areas, while on a harvest control it reported supporting loss in 1 of 3, which is a false support. Live retrieval over the deployed knowledge base returned a relevant chunk for only 2 of 13 answerable questions. Both live results argue for keeping satellite and retrieved evidence advisory. Live language-model evaluations were blocked by a free-tier quota. *None of these results is a measurement on real claims; they show rule enforcement and engineering feasibility, not accuracy.

***Index Terms*—crop insurance, claim verification, remote sensing, Sentinel-2, NDVI, evidence fusion, retrieval-augmented generation, agricultural advisory, auditability, Tamil**

I. Introduction

Indemnity and index crop insurance depend on knowing whether a loss occurred, on which land, and who may claim it. Remote sensing is used to design area indices, to monitor parcels and to smart-sample crop-cutting experiments [1]–[3], and India's YES-TECH programme blends modelled and crop-cutting yields at insurance-unit level [4]. Basis risk remains, and resolution and timing affect it [5]–[7]. Picture-based insurance has been proposed as a farmer-level complement [8]. Trust is a barrier to insurance take-up in India [9], and duplicate insurance of the same land has been reported in the press [10]. At the same time, language and vision-language models can read documents and photographs [11], [12], but hallucination is a documented risk [13], and for high-stakes decisions interpretable rules and audit trails are preferred to opaque models [14], [15].

This paper describes a system built on that premise: models produce *evidence*; a deterministic engine produces the *decision*; and every decision is tied to a versioned policy and a snapshot of the evidence it used. Our contributions are:

1) A verification design in which each evidence source maps to an explicit per-component outcome, provider failure and missing evidence are kept distinct from negative findings, and a required-component gate can only make an approval more cautious, holding the claim rather than deciding when evidence is pending.
2) An open implementation: a farmer-facing web application and backend with context-sufficiency logic for bilingual chat, parcel-geometry validation, cross-farmer overlap as a review signal, Tamil/Latin name matching, and a pre/post-event Sentinel-2 NDVI step on the claim polygon with scene-classification masking.
3) A conformance evaluation on synthetic benchmarks and a live feasibility study (satellite, retrieval, end-to-end farmer journeys), with every post-hoc change disclosed, and a protocol for the real-data experiments that remain.

We do not claim novelty of the individual techniques, which have precedent, nor any accuracy on real claims.

II. Related Work and Research Gap

Satellite crop-damage assessment. Sentinel-2 [16] and NDVI [17] are standard for vegetation monitoring. Pre/post-event NDVI differencing has mapped flood damage to crops, with the authors noting the lack of direct ground validation [18]; Sentinel-2 has been used for hail [19] and flood damage with in-situ data [20], and for land-use anomaly checks in EU subsidy monitoring [21]. Closest to our decision logic, a rule-based decision matrix fusing Sentinel-1 and Sentinel-2 evidence mapped jute lodging after Cyclone Amphan in India [22]. Cloud and shadow masking needs validation [23]–[25], and Tamil Nadu has a public field-level crop dataset [26]. Satellite yield estimation can track ground measures on smallholder plots [27], [28].

Agricultural assistants and grounding. Retrieval-augmented generation [29], [30] has been applied to agriculture with domain constraints [31], compared with fine-tuning [32], and deployed as personalised farmer chat with LLM-judged faithfulness [33]. Agricultural question-answering benchmarks and systems include AgXQA [34] and

KisanQRS [35]; AgriLLM studied Kisan Call Centre queries from Tamil Nadu [36]. Whether retrieved context is *sufficient* to answer has been proposed as a lens on retrieval failures [37]; automated RAG metrics exist [38]. Voice advisory in India has been evaluated [39], and vision-language models remain weak on fine-grained agricultural recognition [40].

Gap. In a search of 74 verified sources (16 October 2026) we found rule-based fusion of satellite evidence for damage mapping [22] and farmer-level photograph insurance [8], but no work that gates individual crop-loss claims on claimant-rights, geometry, photograph, document, satellite and weather components with explicit failure semantics, treats cross-farmer polygon overlap as a review signal, or matches Tamil/Latin names on land documents; nor a public labelled dataset of Indian claim outcomes. Absence in a narrow search is weak evidence, so we make no priority claim.

III. System Design

The backend is a Node.js service on AWS Lambda with MongoDB; the farmer-facing client is a React web application in English and Tamil. A farmer registers parcels as polygons; a claim is made against a registered parcel with a drawn affected area and a declared relationship to the land. The server computes area, validates geometry, and never accepts area, state or decision from the client.

A. Evidence components and outcomes

Ten components describe a claim: parcel identity, claimant rights, land documents, geometry, crop, satellite, photographs, weather, overlap and manual review. Each resolves to one of seven outcomes. A provider that errors or is disabled yields *unavailable*; missing evidence yields *unverified*; *failed* requires a positive finding that contradicts the claim. Evidence derived by a model (photographs, document reading) is never authoritative, and a model reading alone can never verify claimant rights. Satellite and weather are supporting evidence and are never *failed*. Evidence is append-only: a re-run with new inputs adds a row, and the newest row per component is current.

B. Deterministic decision and the gate

A frozen rule engine applies, in order: claim window, event type, area allowance, containment in the parcel, duplicate or partial overlap with the farmer's own claims, remaining eligible area, and the photograph assessment. A versioned policy then names the required components. An approval whose required component is unverified or failed becomes a request for more evidence; one whose required component is pending, unavailable, inconclusive or needs review is *held*: the claim stays submitted and withdrawable, and no decision is recorded. The gate never produces a rejection; rejection stays with the frozen rules and an audited human override. Each decision stores the policy identifier and a hash of its content, the component outcomes and references to the evidence rows used. Slow evidence steps run from a database-backed job queue with leases, so a Lambda that is frozen mid-step cannot lose work.

C. Geometry and overlap

Polygons are validated (closed ring, finite in-range coordinates, no self-intersection, non-zero area). Containment uses a tolerance. Overlap with another farmer's claim is found through a bounding-box pre-filter and an exact intersection, and is a review signal only, since shared boundaries, joint holdings and cadastral error are legitimate causes.

D. Documents and claimant relationship

A vision-language model reads a land document into structured fields; the document is untrusted data, and instruction-like text inside it is itself a finding. Holder names are compared with the claimant across Tamil and Latin scripts by a consonant-skeleton matcher. Each declared relationship (owner, joint owner, tenant, authorised cultivator, heir) has its own expected document set. No authorised Tamil Nadu land-record API is available to us; the official-record adapter is provider-neutral and reports *unavailable*.

E. Satellite evidence

For each claim the system requests per-pixel NDVI statistics over the claim polygon from the Copernicus Data Space Statistical API [41] for a pre-event window (event -40 to -5 days) and a post-event window ($+1$ to $+20$ days, capped at the present), masking pixels the Sentinel-2 L2A scene classification marks as no-data, saturated, cloud shadow, cloud or cirrus. Water is deliberately not masked, so flooding stays observable, which is also a confound for paddy. A drop in mean NDVI of at least 0.15 with sufficient valid pixels in both windows is reported as *supporting loss*; otherwise *no change* or *inconclusive*. The thresholds are provisional and uncalibrated. A change in a vegetation index does not identify a cause. Historical rainfall [42] is recorded as supporting evidence only.

F. Profile-aware advisory chat

For each message the assistant classifies intent, loads only the context that intent needs from the farmer's own records (optionally a selected parcel), tags each item with its provenance (user-provided, verified, third-party, inferred, missing, stale), and asks a follow-up only when a required slot cannot be resolved, about exactly one thing. Retrieved knowledge is fenced and labelled as data, and its source files are returned to the client. Replies follow the requested or detected language.

IV. Evaluation Design

Conformance benchmarks. Deterministic, seeded and reproducible from the repository (`npm run eval`). Ground truth is analytic (geometry), the design rule (fusion), or a written specification (chat); the name set is author-labelled. No network or real data are used.

Live feasibility study (16 October 2026). (i) *Satellite:* the production adapter, unchanged, on 100 m squares near three documented events, south Tamil Nadu floods of 17–18 December 2023, Cyclone Michaung of 4 December 2023 and Cyclone Fengal of 30 November–1 December 2024, located

TABLE I
GEOMETRY CHECKS AGAINST ANALYTIC GROUND TRUTH (SYNTHETIC RECTANGLES PLACED ACROSS TAMIL NADU).

Check	n	Precision	Recall
Outside-parcel detection	300	1	1
Cross-farmer overlap flag	300	1	1
Invalid-geometry detection	190	1	1

Median relative area difference from an ellipsoidal reference: 0.63% (systematic).

TABLE II
FOLLOW-UP QUESTION DECISIONS, SCORED AGAINST A WRITTEN SPECIFICATION (55 SYNTHETIC CASES: 44 WHERE NO QUESTION IS NEEDED, 11 WHERE ONE IS).

	Previous rule	Context logic
Unnecessary questions asked	81.8% (36/44)	0.0% (0/44)
Needed questions missed	27.3% (3/11)	0.0% (0/11)
Crop assumed when ambiguous	100.0% (6/6)	0.0% (0/6)

Intent classification accuracy 100.0% (29/29). Context assembly latency p50/p95 0.37/0.75 ms with weather stubbed.

TABLE III
EVIDENCE FUSION OVER A 192-SCENARIO FACTORIAL: INVALID APPROVALS, CLAIMS REJECTED ON THE AI READING ALONE, AND CLAIMS HELD FOR REVIEW.

Policy	Invalid approvals	Rejected on AI alone	Held
photo-only (v1-legacy)	10/190	2/2	0
full fusion (v2-components)	0/190	0/2	40
v2 without claimant_rights	4/190	0/2	42
v2 without overlap	2/190	0/2	38
v2 without photos	0/190	0/2	40
v2 without geometry	0/190	0/2	40

TABLE IV
SIMULATED BEHAVIOUR OF THE PROVISIONAL NDVI-CHANGE RULE (NO REAL IMAGERY; 400 TRIALS PER SETTING).

Setting	Outcome rate
No change, clear sky: flagged as supporting loss	0.0%
No change, cloudy: flagged as supporting loss	0.0%
NDVI drop 0.15, clear sky: detected	49.8%
NDVI drop 0.30, clear sky: detected	99.8%
Drop 0.30, 70% of pixels masked: inconclusive	100.0%

from humanitarian situation reports, and on a Thanjavur control dated 15 February 2024, within the samba harvest season, with three squares each. Land cover, crop and actual damage at each square were not verified, so these are case studies, not labelled data. (ii) *Retrieval:* 22 queries (13 answerable from the deployed corpus, 9 advisory) against the production collection, read-only, with relevance by a mechanical rule fixed before the run. (iii) *Acceptance:* the application's farmer journey (profile, parcel, chat, claim, photographs, land document, submission, verification, cross-user access) run over HTTP against the real application on a local database with mock storage, under both policies. (iv) Live chat and synthetic land-document extraction were prepared but could not run (Section VII).

V. RESULTS

Geometry (Table I). Containment, overlap flagging and invalid-geometry detection agree with analytic geometry. The harness exposed two issues: the shared validator accepted out-of-range coordinates (now rejected), and the server's area is about 0.6% larger than an ellipsoidal reference at Tamil Nadu latitudes because the formula uses the equatorial radius; we report rather than change it, so stored areas stay comparable.

Evidence fusion (Table III). The legacy photograph-and-geometry policy approves claims with missing, pending or inconsistent ownership evidence and with overlap on another farmer's claim; the gate does not. Satellite and weather outcomes never changed a decision in 1,536 comparisons per policy, by design. Because the label is the design rule, this shows rule enforcement, not accuracy.

Chat context (Table II). Against the previous rule, which asked a follow-up whenever a message ended with a question mark, the context logic asked no unnecessary questions and missed none on the specification. The specification and logic share an author, and the baseline is a reimplementation.

Satellite, simulated (Table IV) and live (Table V). All 12 live runs completed in a median of 2.5 s (maximum 11.8 s). Near the three events the rule reported supporting loss in none of 9 squares: 5 showed no change, 1 had too few valid pre-event pixels, and 3 had no usable cloud-free post-event observation in the window. Three of the analysed event squares had a pre-event NDVI between -0.04 and 0.09, consistent with bare, flooded or recently transplanted fields, so a fixed window cannot observe a loss there and NDVI rose afterwards. On the harvest control one square reported supporting loss ($\Delta\text{NDVI} = -0.25$, from 0.74 to 0.49 across the harvest), which is a false support produced by normal crop phenology. These observations support the design choice that satellite evidence is optional and never decisive, and they show that a fixed window needs crop-calendar context before thresholds can be calibrated.

Retrieval (Table VI). The deployed corpus (450 chunks from 60 files) consists of district statistics tables and a geography file, with no advisory text, so no advisory question could be grounded (0 of 27 chunks). For questions the tables can answer, the right file was retrieved for 10 of 13, but in 9 of 11 such hits the chunk was a header without data rows; a relevant chunk was retrieved for only 2 of 13. The cause is in ingestion: rows are chunked without the file name or crop, so a district's row is rarely retrieved with its table. Source attribution therefore matters: a farmer and an auditor can see that an answer was not grounded.

Acceptance. Under both policies every journey step except live chat passed (24 of 26 under the legacy policy, 25 of 27 under the gated policy), including idempotent replay, cross-user access returning "not found", and a client-supplied verification result being ignored. Under the legacy policy the claim was approved on photographs and geometry with claimant rights unverified; under the gated policy it was returned for more evidence, and after a land record was attached and the claim resubmitted it was held with claimant rights pending review, the intended behaviour.

Name matching. On 60 author-labelled pairs the matcher scored 55 of 60 before a rule change and 60 of 60 after; the change followed those errors on the same set, so the second figure is optimistic.

TABLE V
LIVE SENTINEL-2 L2A RUNS OF THE PRODUCTION ADAPTER ON 100 M SQUARES NEAR DOCUMENTED EVENTS AND A HARVEST CONTROL (RUN 2026-10-16). LAND COVER AND CROP AT EACH SQUARE WERE NOT VERIFIED; THE EVENT ROWS ARE CASE STUDIES, NOT LABELLED DAMAGE.

Event (date)	Squares	Supports loss	No change	Inconclusive	ΔNDVI
south_tn_floods_2023 (2023-12-18)	3	0	2	1	0.11, -0.02
cyclone_michaung_2023 (2023-12-04)	3	0	2	1	0.46, 0.31, 0.56
cyclone_fengal_2024 (2024-12-01)	3	0	1	2	0.03
thanjavur_harvest_control_2024 (2024-02-15)	3	1	1	1	-0.25, -0.07, 0.21

TABLE VI
LIVE RETRIEVAL ON THE PRODUCTION COLLECTION (FARMING-DOCUMENTS, EMBED-ENGLISH-V3.0, TOP-3). RELEVANCE BY A MECHANICAL AUTHOR-WRITTEN RULE, NOT HUMAN JUDGEMENT.

Measure	Rate (k/n)
Answerable: any relevant chunk	15.4% (2/13)
Answerable: top-1 relevant	7.7% (1/13)
Answerable: expected file retrieved	76.9% (10/13)
Expected-file hits that are header-only	81.8% (9/11)
Advisory: answer-bearing chunks	0.0% (0/27)

VI. DISCLOSURES: DEFECTS FOUND AND POST-HOC CHANGES

The live study found defects that no synthetic test had exposed. We list them because they bear on how far the conformance results can be trusted.

- *Response parsing.* The production NDVI parser read the provider's statistics at the wrong nesting level, so every live analysis returned "no valid statistics". It was fixed, and a recorded live response is now a regression fixture.
- *Scene cloud filter (post-hoc).* A 30% scene-level cloud limit rejected all monsoon imagery for the event windows. We raised the scene limit used for parcel statistics to 90% after observing this, relying on per-pixel masking for the parcel itself. The NDVI thresholds were not changed.
- *Prototype request.* The earlier proof-of-concept request was rejected by the live service (HTTP 400, missing data-mask output); it had never run live. Production no longer uses it.
- *Quota masked as timeout.* A model-provider quota refusal was retried with back-off until the 25 s timeout fired, so farmers would have seen "did not respond in time". Retries are now capped and the refusal is reported as a temporary unavailability.
- *Stale decision on hold.* A claim held after a resubmission still reported the previous round's decision. The hold now clears it; the earlier decision remains on the audit trail.
- *Write budget.* A realistic single-claim journey exceeds the per-user limit of five claim writes per hour. We report this rather than change the production limit.

VII. BLOCKED EVALUATIONS

The configured language-model key was on a free tier limited to 20 requests per day for the model used, and the limit was reached during the study. The live chat comparison (context-aware versus generic, 46 items) and synthetic land-document extraction (12 rendered, watermarked documents, 3 with injected instructions) are implemented and ready to run but produced no results, and we report none. Latency, token and cost figures for chat are therefore absent.

VIII. DISCUSSION

The structural finding is that separating evidence from decision, and failure from negative finding, makes behaviour under outages and missing data explicit and testable, and lets a stricter policy be introduced without changing the frozen rules. The live study adds two empirical cautions that the synthetic study could not: a fixed-window NDVI change rule is confounded by crop phenology and sparse post-event observation, and retrieval quality depends on corpus content and chunking more than on the retriever. Both favour the conservative roles the design assigns to these sources.

IX. LIMITATIONS AND THREATS TO VALIDITY

- No real claims, farmers or documents were used. No accuracy, fraud-detection or loss-estimation figure is claimed.
- Live satellite cases are 12 squares chosen near documented events without verifying land cover, crop or damage; they are not labelled data, and one parameter was changed after observation.
- Specifications and relevance rules are author-defined; the name matcher was tuned on its own evaluation set.
- Live language-model behaviour is unmeasured. The deployed knowledge base contains no advisory text.
- No authorised land-record API is available; the required-document policy per relationship is unreviewed by any legal or insurer authority.

X. ETHICAL, LEGAL AND PRIVACY CONSIDERATIONS

No real farmer data were used; the synthetic documents are watermarked and use fictional names. Real use needs informed consent, secure handling of documents, permission for any official record and human review of adverse and uncertain outcomes; under the gated policy no model reading alone rejects a claim. The statement in the repository has not been reviewed by a lawyer or ethics board.

XI. CONCLUSION AND FUTURE WORK

We described, implemented and tested an evidence-gated claim-verification design and a profile-aware advisory chat, and ran the satellite, retrieval and farmer-journey paths live. The results show that the components enforce their specifications and that the live paths work; they do not show real-world accuracy. Next steps are: live chat answers rated

by agronomists in English and Tamil; labelled document extraction; satellite validation against independently labelled parcels with crop-calendar-aware windows and Sentinel-1 for cloudy periods; re-chunking the knowledge base with advisory sources; and evaluation on authorised cadastral data.

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